Muti-Head Attention is the idea of instead of running attention once per layer, it runs several times in parallel. Each sub-run is called a head.

A head = one full attention cycle with different weight matrices to focus on different kinds of relationships.

The core problem it tackles

A single head = one softmax = one weighted average.

So all of a word's relationships gets fused and mixed into a single blend and 512 numbers, which can't be separated

The problem isn't the too few dimensions (512) but only one weighting and single average.

The solution

- The vector is split into 8 sections of 64 dimensions.
- Each head runs its own attention cycle and softmax, to weight things differently.
- As weightings are separate, information of different words are stored side-by-side rather than on top of each other.
- The result is multiple independent weightings instead of one in the same 512 dimensions.
- The 8 heads are then concatenated back to 512 dimensions.
- It's then multiplied by the matrix W^o to allow all of the head outputs communicate with each other.
- The output is a vector with distinct sections of understanding.

Matrix W^o

- W^o is multiplied by the concatenated vector.
- W^o allows each head to pull relevent information from other heads.
- W^o is obtained via training

